

The Blind Spot Paradox

When a self-repairing model destroys the evidence its own monitor needs

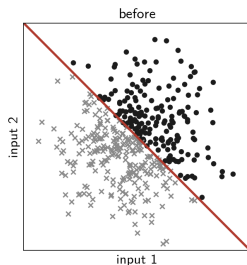
Alexandre Boyer Melaine Gouillou Salomé Fonvielle Ulysse Petit-Tichanné

Research Track, Topic 39, CentraleSupélec. Supervisor: Raphaël Minato

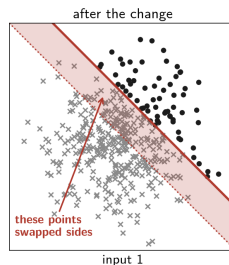
September 2026

Concept drift: the rule changes, the data does not

$P(X)$ unchanged
the data



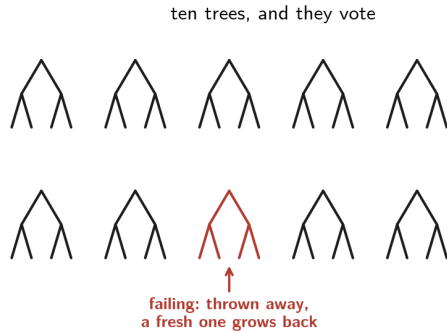
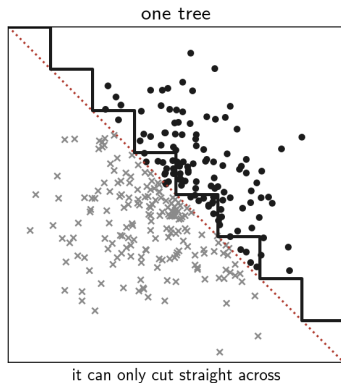
$P(Y | X)$ changes
the rule



one dot = one observation; its shape is the answer the model has to predict

A model trained on the left keeps predicting, confidently, and is now **wrong on the shaded band**.

Defence one: a model that repairs itself

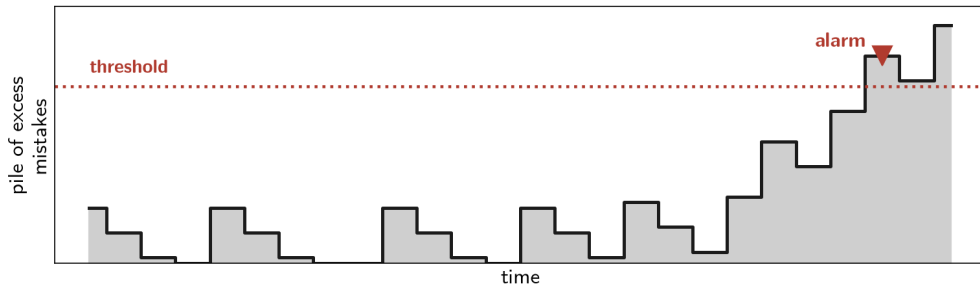


This is the **Adaptive Random Forest**. Ten trees, replaced one at a time.

Defence two: a watchdog that raises the alarm

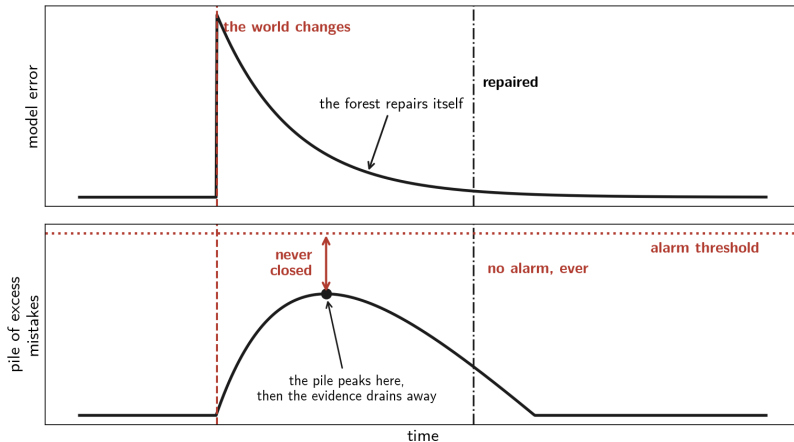
the model answers, one observation at a time (x = wrong)

x o o o x o o o o x o o o x o o x x o x x o x



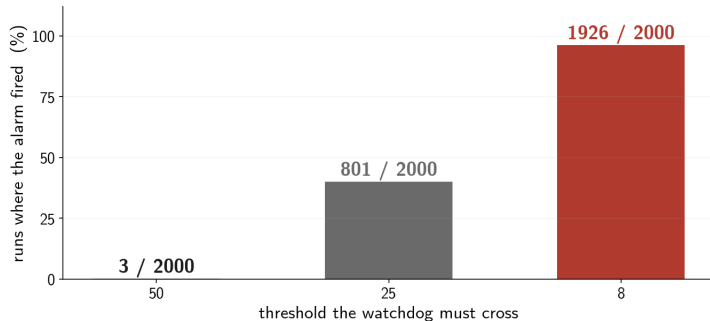
Two communities, two tools. What happens when both run at once?

Put them together, and they cancel out



The repair has a date. The alarm never gets one. (schematic)

Why not just lower the threshold?

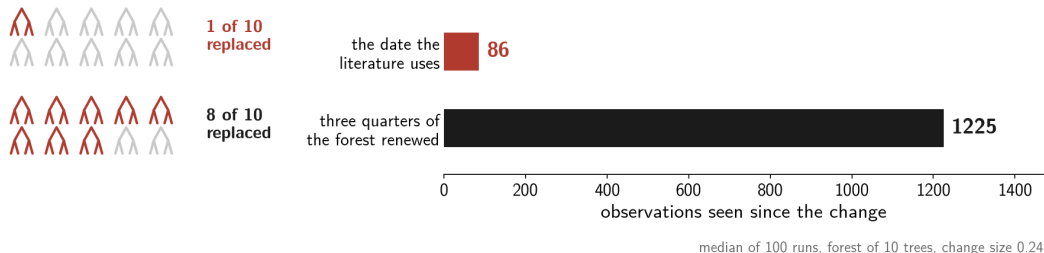


2 000 runs per threshold, same drifts, same seeds

A low threshold also fires when **nothing has happened**, and an alarm that cries wolf gets switched off.

Our contribution: when is the forest actually repaired?

To call the race, you have to **time the repair**.



It fires at the *first* tree. It times the start of a repair, not its end.

Measuring the repair in mistakes, not in trees

Measure	What it counts	Value
First tree replaced τ_{ARF}	the date the paper uses	86
Error back to normal τ_{rec}	the error settles back to its usual level	261
Forest 3/4 renewed τ_{swap}	8 trees out of 10 replaced	1 225
Excess mistakes $A(H)$	the extra errors the change caused	38
Watchdog's peak $S_{\text{max}}(H)$	the highest point of the watchdog's pile	33

The paper's date barely tracks the damage.

Same change as the previous slide, median of 100 runs. First three: observations after the change. Excess mistakes: counted over the next 2 000 observations. Watchdog thresholds: 8, 25 and 50.

Where we stand

The blind spot is real

At the paper's strict threshold, the alarm fired **3 times in 2 000** runs.

Repair dated too early

The paper times it at the **first tree** replaced: up to **14×** too early.

Open question

What should replace the first-tree date? Candidates under test, none has passed yet.

Thank you for your attention.